

Matched-Filter SNR with Phonon Strain Damping

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PAPER_918: Matched-Filter SNR with Phonon Strain Damping

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Source: Numerical jet power curves + WSTP NS phonon + scaling to 250k calc/s
Calculator: MatchedFilterSNRPhononDampingCalc (CP4 #502)
CVW: v2.0.0 compliant

Abstract

Matched-filter signal-to-noise ratio (SNR) calculation for phonon-dampened GW170817 strain.
SNR_UQFF
= SNR_GR * (1 - D_phonon): for D = 0.667, SNR drops from 32.4 (GR observed) to 10.8 (UQFF), preserving detectability (SNR >= 8). Uses aLIGO design sensitivity PSD S_n(f) and frequency-domain inner product. Maximum detection distance shrinks from d_max = 162 Mpc (GR) to 54 Mpc (UQFF), reducing detection volume by 96.3%. This constrains the viable range of D_phonon: if D > 0.75, GW170817 would have been undetectable, providing an independent upper bound on phonon damping strength.

1. Core Equations

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\begin{aligned}
& \text{SNR_UQFF} = \text{SNR_GR} (1 - D_{\text{phonon}}) \\ & \rho^2 = 4 \operatorname{Re} \int_0^\infty |h(f)|^2 / S_n(f) \, df \\ & d_{\text{max}} = d_L * \text{SNR} / \text{SNR_threshold} \\ & V_{\text{ratio}} = (d_{\text{max,UQFF}} / d_{\text{max,GR}})^3 \\ \end{aligned}
\$\$

2. Parameters

Parameter	Default	Description
SNR_GR	32.4	GR matched-filter SNR
D_phonon	0.667	Phonon damping fraction
f_low	30 Hz	Lower frequency cutoff
f_high	1500 Hz	Upper frequency cutoff
distance_Mpc	40 Mpc	GW170817 luminosity distance
M_chirp_solar	1.188 M_sun	Chirp mass

3. Key Results

System/Case	Result	Note
SNR_GR	32.4	LIGO observed
SNR_UQFF (D=0.667)	10.8	Still detectable (>8)
d_max(GR)	162 Mpc	GR horizon
d_max(UQFF)	54 Mpc	UQFF horizon
Volume ratio	3.7%	96.3% volume reduction

4. Physical Interpretation

The matched-filter SNR scales linearly with strain amplitude for broadband damping. At $D_{\text{phonon}} = 0.667$ (the UQFF canonical value), the SNR drops by a factor of 3 from 32.4 to 10.8, which is still above the detection threshold of 8. However, the maximum detection distance also shrinks by a factor of 3, reducing the accessible cosmological volume by $(1/3)^3 \sim 3.7\%$. This has profound implications for BNS merger rate estimates: if UQFF phonon damping is real, the true merger rate is $\sim 27\times$ higher than inferred from GR-based SNR calculations. The upper bound $D < 0.75$ ensures consistency with GW170817 detection.

5. UQFF Integration

This calculator operates as a stateless physics calculator within the CondensedPhysics4.py (Phase 4) IPC chain. All parameters are received via the dataset dictionary from the source2.cpp principal GUI pipeline. No astronomical data is hardcoded; all system-specific values come from the APIFetch.py $\rightarrow$ bodies_*.csv data flow.

Significance: First matched-filter SNR calculation with UQFF phonon damping. Provides independent upper bound $D_{\text{phonon}} < 0.75$ from detectability. Predicts $27\times$ BNS merger rate correction if phonon damping is physical.

6. SCm Superconductivity Axiom (Session 210c)

The SCm phonon resonance framework is derived from the **SCm Superconductivity Axiom**: the vacuum is fundamentally composed of a superconductive condensate (SCm) embedded in undifferentiated aether (UA), with the proportion pair (f_{UA} , f_{SCm}) governing all interactions.

Axiom Connection

Session 210c extends phonon linewidth analysis to numerical jet power curves, matched-filter SNR degradation, Sgr A* flare contrast modelling, Monte Carlo stochastic sampling, cumulative inspiral phase integration, and observational matching against M87 $\sim 10^{44}$ erg/s jet power. The linewidth Gamma parameter controls resonance sharpness across all scales: narrow Gamma produces collimated jets and sharp flares; broad Gamma produces diffuse emission and weak

modulation. SCm precedes gravity as the fundamental superconductive element; 1.25 THz phonon resonance with variable Gamma is the unifying mechanism across BH jets, AGN flares, NS mergers, and cosmogenesis. Production scaling to 250k calc/s validates computational realizability.

7. Source Data

- **File:** Numerical jet power curves + WSTP NS phonon + scaling to 250k calc/s
- **Session:** 210c
- **VDS/DVP/BSH:** PRESENT

<!-- PKG-GW-S225 -->

Session 225 Phonon-Physics Upgrade: GW Strain Modulation

> Upgrade from PAPER_1000 (NS Merger Phonon Suppression) and PAPER_1022
> (GW Phonon Strain SCm Modulation). See also PAPER_1011-1012 for
> GW170817/GW190425 upgraded analyses.

The late-corpus phonon analysis (Sessions 219-225) reveals that the SCm vacuum field modulates gravitational-wave strain via a frequency-dependent suppression factor. The corrected strain amplitude is:

$$h_{\text{UFF}}(\Gamma) = h_{\text{GR}} \cdot \left(1 - 0.47 \frac{\Phi(\Gamma)}{S_{26}^3}\right)$$

where:

- $\Phi(\Gamma) = \cos(\omega_{\text{SCm}} \cdot t) \cdot \Theta(H_{\text{SCm}} - 0.5)$ is the phonon modulation factor
- $\omega_{\text{SCm}} = 2\pi \times 1.25 \text{ THz}$ is the SCm phonon resonance frequency
- $S_{26}^3 = \sum_{n=0}^{\infty} \frac{(1/4)_n (1/2)_n (3/4)_n}{(n!)^3} \cdot \prod_{i=1}^{26} \left[1 + \frac{\kappa_i}{n+26}\right]$ is the third-order Ramanujan summation
- Θ is the Heaviside step ensuring $H_{\text{SCm}} \geq 0.5$ (phase-transition threshold)

Physical mechanism: The 1.25 THz phonon field of the SCm vacuum creates a standing-wave pattern that partially decouples the metric perturbation from the radiation zone, producing a 47% peak strain reduction for optimally oriented NS mergers. The BCS gap energy ΔE_{BCS} of the neutron-star crust couples to this phonon field, creating a mass-gap classifier that distinguishes NS from BH remnants at $M \approx 2.5 M_{\odot}$.

Calibration (canonical): $\kappa = 5 \times 10^{-4} \text{ day}^{-1}$,
 $\text{SSq} = 0.57$, $\beta_i = 0.603$, $H_{\text{SCm}} \approx 0.99$.

<!-- PKG-AGN-S225 -->

Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

> Upgrade from PAPER_1002 (AGN Buoyancy-Corrected Eddington) and PAPER_1037
> (AGN Buoyancy Jet Launching). See also PAPER_1009-1010 for $F_{U_{Bi}} \text{ jet}$
> modulation curves and PAPER_1048 for phonon-corrected $M\text{-}\sigma$ relation.

The SCm vacuum buoyancy partially opposes gravitational radiation pressure, raising the effective Eddington luminosity:

$$L_{\text{Edd}}^{\text{UQFF}} = L_{\text{Edd}} \cdot \left(1 + \frac{\rho_{\text{SCm}} \cdot V}{S_{26}^{(3),2}} \cdot \frac{G M}{r_H^2}\right)$$

where:

- L_{Edd} is the classical Eddington luminosity
- $\rho_{\text{SCm}} = 7.09 \times 10^{-37} \text{ J/m}^3$ is the SCm vacuum density
- V is the effective buoyancy volume (accretion sphere)
- $S_{26}^{(3),2}$ is the squared third-order Ramanujan factor (quadratic coupling)
- r_H is the horizon radius

Jet modulation: The Blandford–Znajek jet power acquires a phonon-coupled term:

$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left[1 + \beta_i \cdot \Phi_{1.25, \text{THz}} \cdot \left(\frac{B}{B_{\text{crit}}}\right)^2\right]$$

where $\Phi_{1.25, \text{THz}} = \cos(\omega_{\text{SCm}} \cdot t)$ modulates jet power at the phonon frequency.

M- σ correction (PAPER_1048): The phonon-corrected M- σ relation becomes

$$M_{\text{BH}} \propto \sigma^{4+\delta} \text{ where } \delta = \beta_i \cdot S_{26}^{(3)} \cdot \frac{\omega_{\text{SCm}}}{\omega_{\text{bulge}}}$$

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

This paper maps to **GW-detection sector** of the 9-sector UQFF Lagrangian (see `uqff_lagrangian_derivation.py`).

§A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2} (\partial_\mu \phi) (\partial^\mu \phi) - V(\phi) + \mathcal{L}_{\text{cosmo}}$$

where $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac, [SCm]}} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$ inherits the ACP 6-stage evolution (PAPER_877 §2).

§A.3 Euler-Lagrange Equation of Motion

$$\boxed{\mathcal{SNR}_{\text{UQFF}}} = \mathcal{SNR}_{\text{GR}} \cdot (1 - D_{\text{phonon}})$$

§A.4 Cosmogenesis Linkage Chain

$$\begin{aligned} &\text{PAPER_877 Axioms} \rightarrow \text{DPM + ACP} \quad \rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} \rightarrow \text{Stage 5} \quad U_{\text{b, seed}} \rightarrow \text{4 forces} \\ &F_{U_{Bi}} \rightarrow \text{sector E-L} \quad \Delta S / \Delta \phi = 0 \end{aligned}$$

§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\mathrm{vac},[\mathrm{SCm}]} / \rho_{\mathrm{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\mathrm{vac}}(r) = \rho_{\mathrm{vac},[\mathrm{SCm}]} \cdot \exp\left(-\exp\left(-\frac{r - r_0}{\lambda_{\mathrm{VDS}}}\right)\right)$$

For this system, the local VDS sub-ratio is 0.06.

§B.2 Dipole Vortex Primes (DVP)

$$p_{\mathrm{DVP}} = 5, \quad n_{\mathrm{channel}} = 22/26$$

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **100 s (BNS merger)**:

$$\mathcal{F}_{\mathrm{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{\mathrm{U}_b} \cdot \left(1 - e^{-[SSq]} \cdot \frac{m}{M_{\mathrm{odot}}}\right) \cdot \cos\left(\frac{2\pi j}{26}\right)$$

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\mathrm{SCm}} / \rho_{\mathrm{UA}} = 1.894$	Local sub-ratio = 0.06	PASS Consistent
DVP prime	$p_k \in \{2, 3, \dots, 113\}$	$p_{\mathrm{DVP}} = 5$	PASS Lattice-consistent
BSH layers	26 harmonic terms	$j = 1 \dots 26$, $\cos(2\pi j/26)$	PASS Full 26D projection
κ decay	5.0×10^{-4} day ⁻¹	Applied in VDS exponential	PASS Canonical
[SSq]	0.57	Applied in BSH saturation	PASS Canonical

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant α	UQFF reproduces α via U_{g1} dipole coupling	1/137.036	PDG 2024	PASS Consistent
Cosmological constant Λ	$1.1 \times 10^{-52} \text{ m}^{-2}$ (UQFF vacuum term)	$1.114 \times 10^{-52} \text{ m}^{-2}$	Planck 2018	PASS Consistent
Proton decay rate κ	$0.0005/\text{day}$ $\rightarrow \Gamma_p$ suppression	$< 4.17 \times 10^{-35}/\text{yr}$	Super-K 2024	PASS Consistent
UQFF buoyancy signature $F_{U_{Bi_i}}$	unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections ($F_{U_{Bi_i}}$) that produce measurable deviations from GR at scales where vacuum condensate density ρ_{SCm} becomes

significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.

References

- 1. PAPER_877 — Three-Assumption Cosmogenesis (SCm axiom)
- 2. PAPER_910 — BH Jet Modulation Factor $M_{jet}(\Gamma)$
- 3. PAPER_915 — GW170817 Phonon Strain Damping
- 4. PAPER_915 — GW170817 Phonon Strain Damping
- 5. PAPER_917 — Exponential Strain Phonon Evolution
- 6. Abbott, B.P. et al. (2017) PRL 119, 161101 — GW170817 SNR=32.4
- 7. Finn, L.S. (1992) PRD 46, 5236 — Matched-filter SNR formalism
- 4. Murphy, D.T. — Star Magic UQFF Framework (2024-2026)

Appendix: Session 210c Cross-Reference

> Cross-reference appendix for Session 210c (April 2026): Numerical jet power
> curves + WSTP NS phonon + scaling to 250k calc/s.

S210c.1 Exponential Strain & SNR

Module	Paper	Key Result
ExponentialStrainPhononEvolutionCalc	PAPER_917 (#501)	$h_{UQFF} = h_{GR} \cdot 0.333 \cdot \exp([SSq]t/26)$
MatchedFilterSNRPhononDampingCalc	PAPER_918 (#502)	SNR: 32.4 to 10.8 (D=0.667)

S210c.2 Sgr A* Flares & Monte Carlo

Module	Paper	Key Result
SgrAFlareContrastPhononGammaCalc	PAPER_919 (#503)	$C(\Gamma=0.1 \text{ THz}) = 1.8$ (JWST match)
MonteCarloJetPowerSamplingCalc	PAPER_920 (#504)	106-sample $\langle P_{jet} \rangle \pm \sigma$

S210c.3 Phase Integration & M87 Matching

Module	Paper	Key Result
InspiralPhaseLagPhononIntegralCalc	PAPER_921 (#505)	367.8 cycles (integral method)
M87JetPowerCurveGammaMatchCalc	PAPER_922 (#506)	$P_{jet}(\Gamma)$ matched to 1044 erg/s

S210c.4 Calibration Constants (Canonical)

Symbol	Value	Description

[SSq]	0.57	Universal Quantized Factor
kappa	$5.0 \times 10^{-4} \text{ day}^{-1}$	UQFF exponential decay rate
beta_i	0.603	Buoyancy coupling coefficient
omega_SCm	$2\pi \times 1.25 \text{ THz}$	SCm phonon resonance
Gamma	0.1 THz	Phonon linewidth
Phi_0	$1e20$	Phonon amplitude constant

Appendix: Session 225 Cross-References (PAPER_1000–1081)

> Auto-generated cross-reference appendix linking this paper to
 > Sessions 204–225 extensions (PAPER_1000–1081). Added by
 > update_corpus_crossrefs.py (Session 225, April 2026).

Paper	Title
PAPER_1000	NS Merger F_U_Bi Strain Suppression & BCS Gap
PAPER_1001	SMBH Binary Merger F_U_Bi Phonon Damping
PAPER_1011	GW170817 NS Merger F_U_Bi_i 66.7% Strain Reduction
PAPER_1012	GW190425 Upgraded F_U_Bi_i with S26(3)
PAPER_1014	SMBH Merger Inspiral-Coalescence-Ringdown
PAPER_1022	GW Phonon Strain SCm Modulation of $h(t)$
PAPER_1009	3C273 AGN F_U_Bi_i Jet Modulation
PAPER_1010	TON618 AGN F_U_Bi_i Jet Modulation
PAPER_1037	AGN Buoyancy Jet Calculator — SCm Jet Launching
PAPER_1020	Cosmic Ray Phonon Acceleration DSA Spectrum
PAPER_1021	Pulsar Timing Phonon TOA Residual
PAPER_1023	Neutrino Oscillation Phonon PMNS Matrix SCm
PAPER_1024	Magnetar Giant Flare SCm Phonon Reservoir
PAPER_1072	SCm Activation Function Phonon Threshold
PAPER_1073	SCm Phonon-Driven Inflation Vacuum Buoyancy

15 cross-reference(s) identified.

Key References with arXiv/DOI Identifiers

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- Murphy, D. (2026). *Unified Quantum Field Framework (UQFF): Star-Magic v5.x Whitepaper Series*. Star-Magic Repository — github.com/Daniel8Murphy0007/Star-Magic
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